RADIANT

GUI DESIGN · SCREEN GALLERY

Spatial · Spectral · Radiometric Sensor Modeling GUI

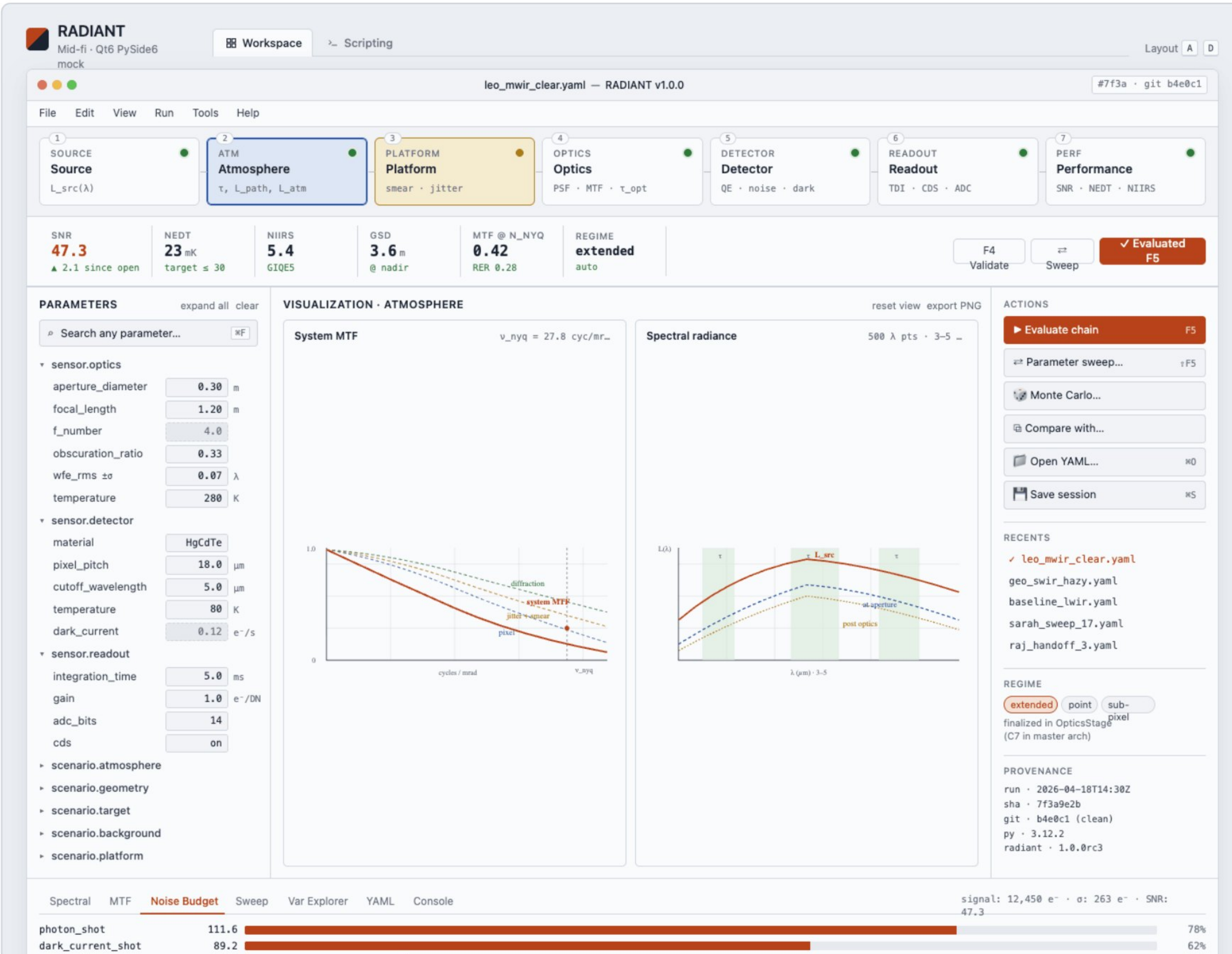
A first-principles electro-optical sensor performance model — propagating spectral radiance through a seven-stage signal chain from source to output.

01 Source | 02 Atmosphere | 03 Platform | 04 Optics | 05 Detector | 06 Readout | 07 Performance

RADIANT v1.0.0rc3 · PySide6 / Qt6 layout study
Workspace · Stage detail · Scripting
Generated June 2026

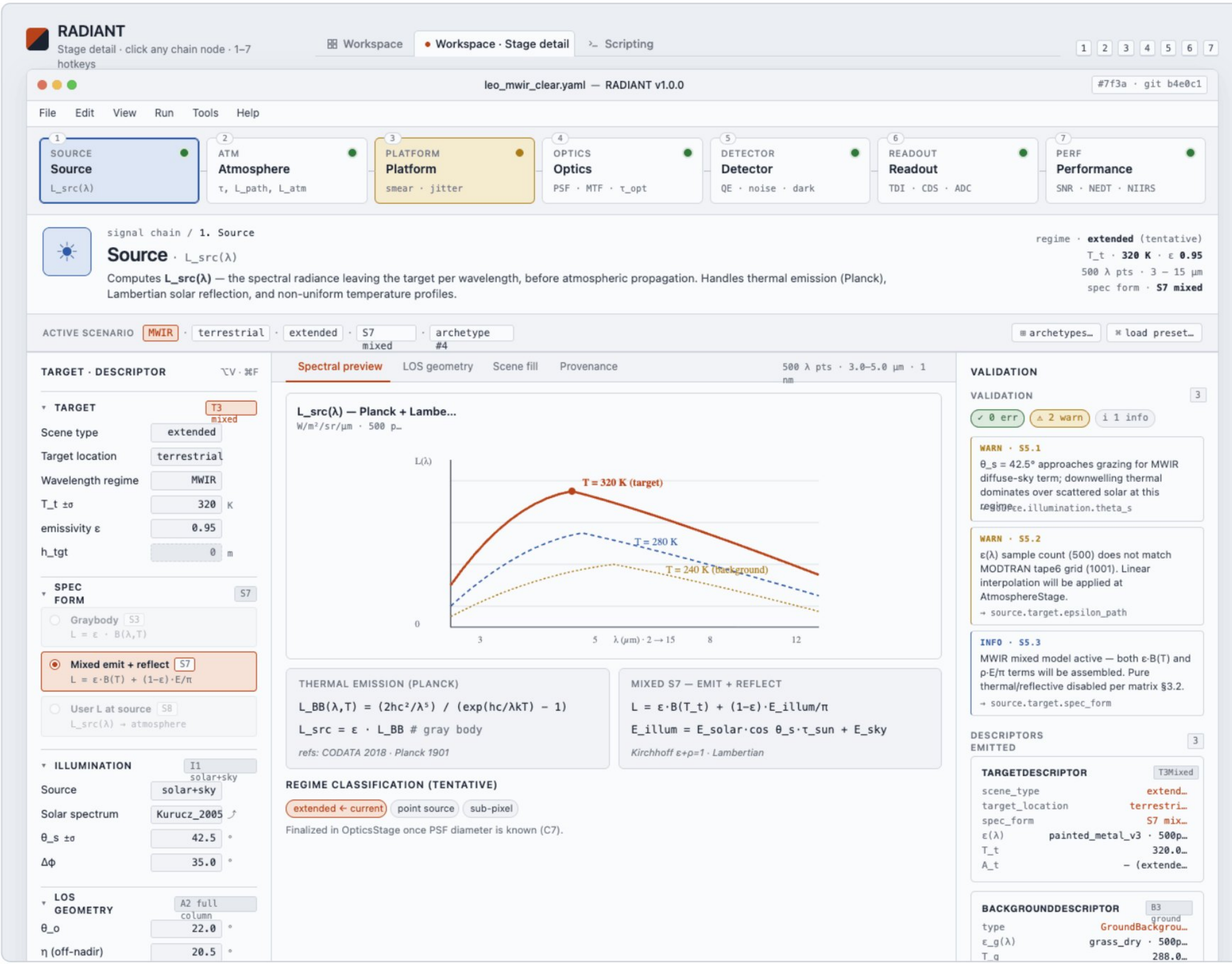
Workspace

The visual home — the seven-stage signal chain with live health indicators, parameter tree, visualization canvas, and per-stage drill-down.



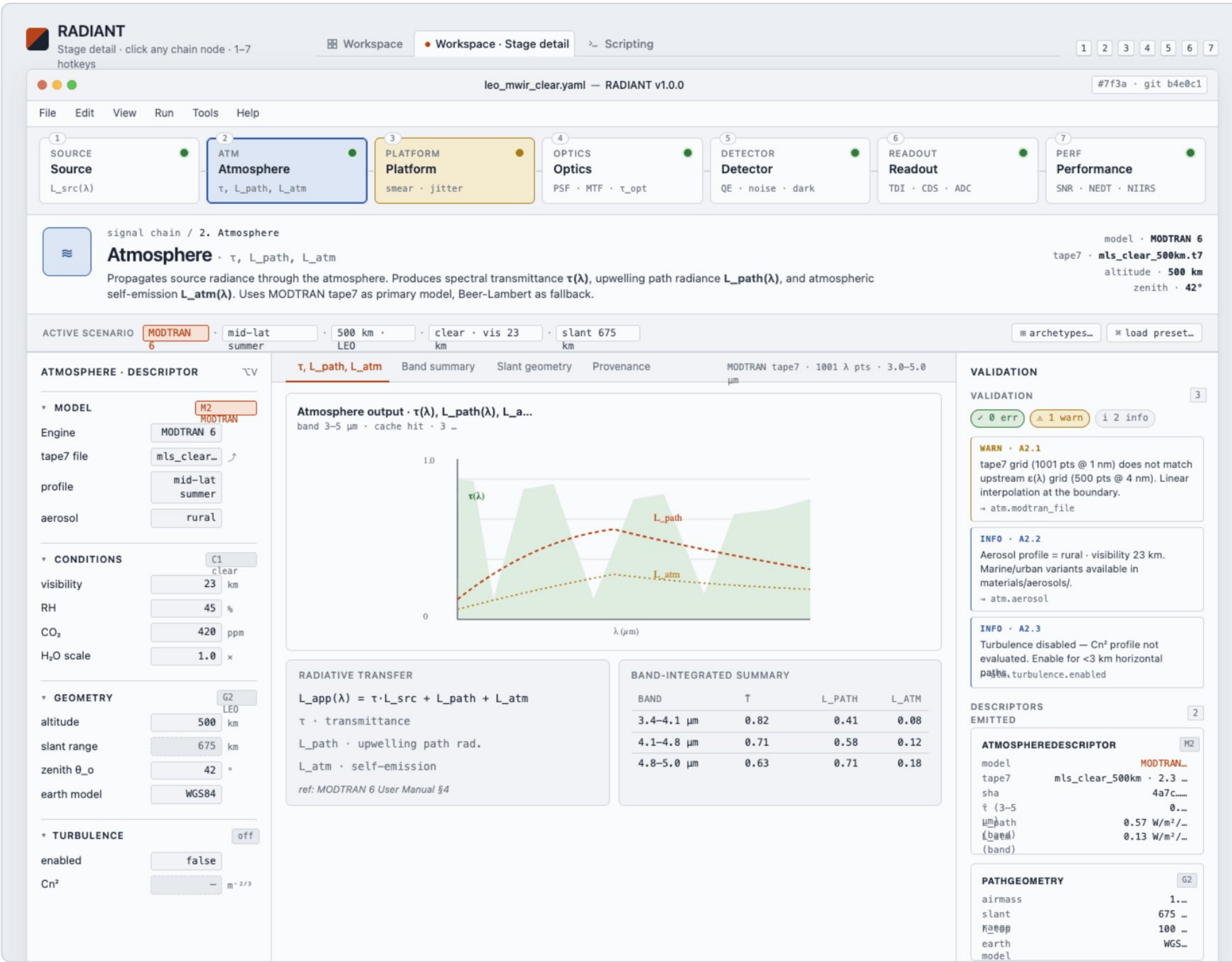
Source

Computes L_src(λ). Target descriptor, spec-form picker (S1–S12), illumination and line-of-sight geometry, with a live validation rail.



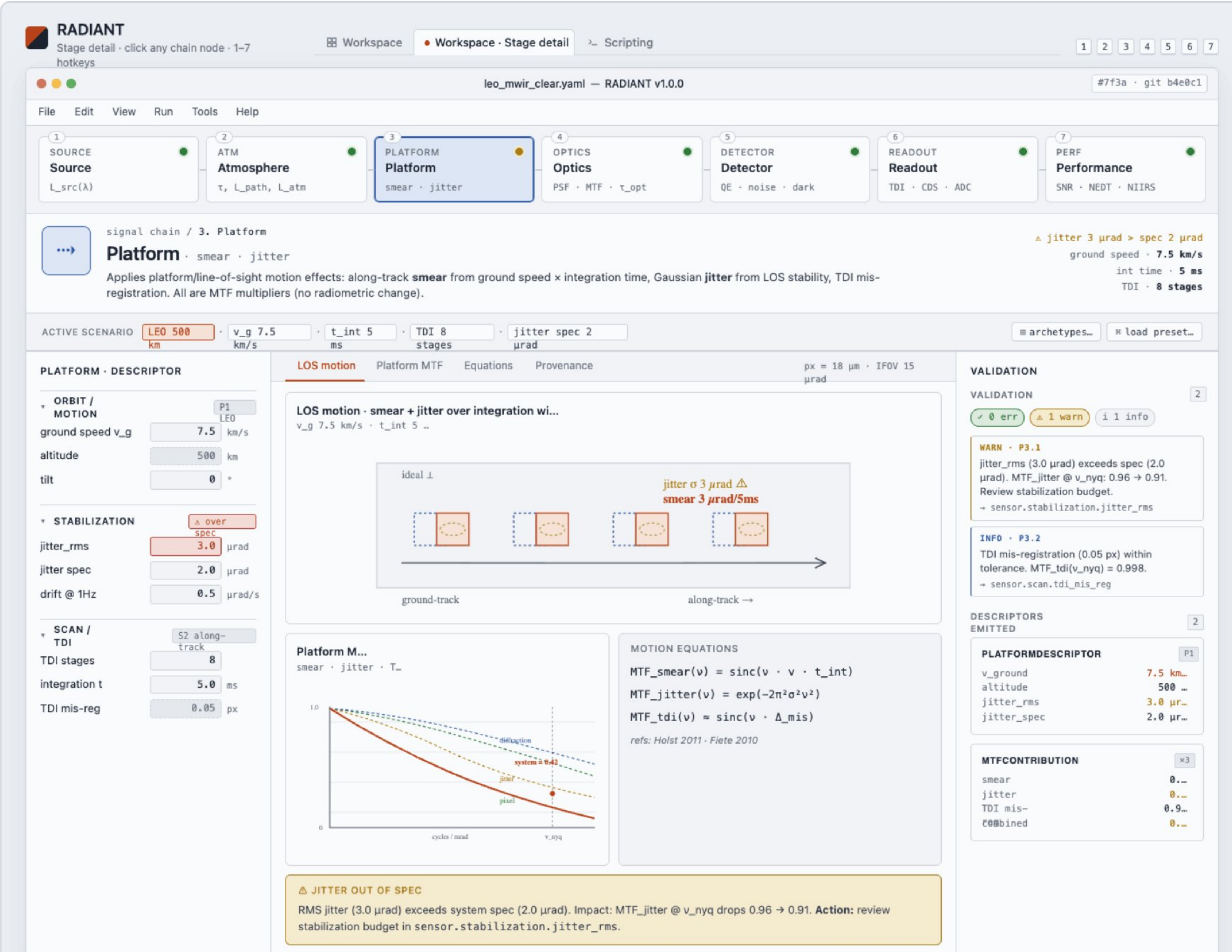
Atmosphere

Propagates source radiance — transmittance τ(λ), path radiance L_path(λ), and self-emission L_atm(λ) via MODTRAN tape7.



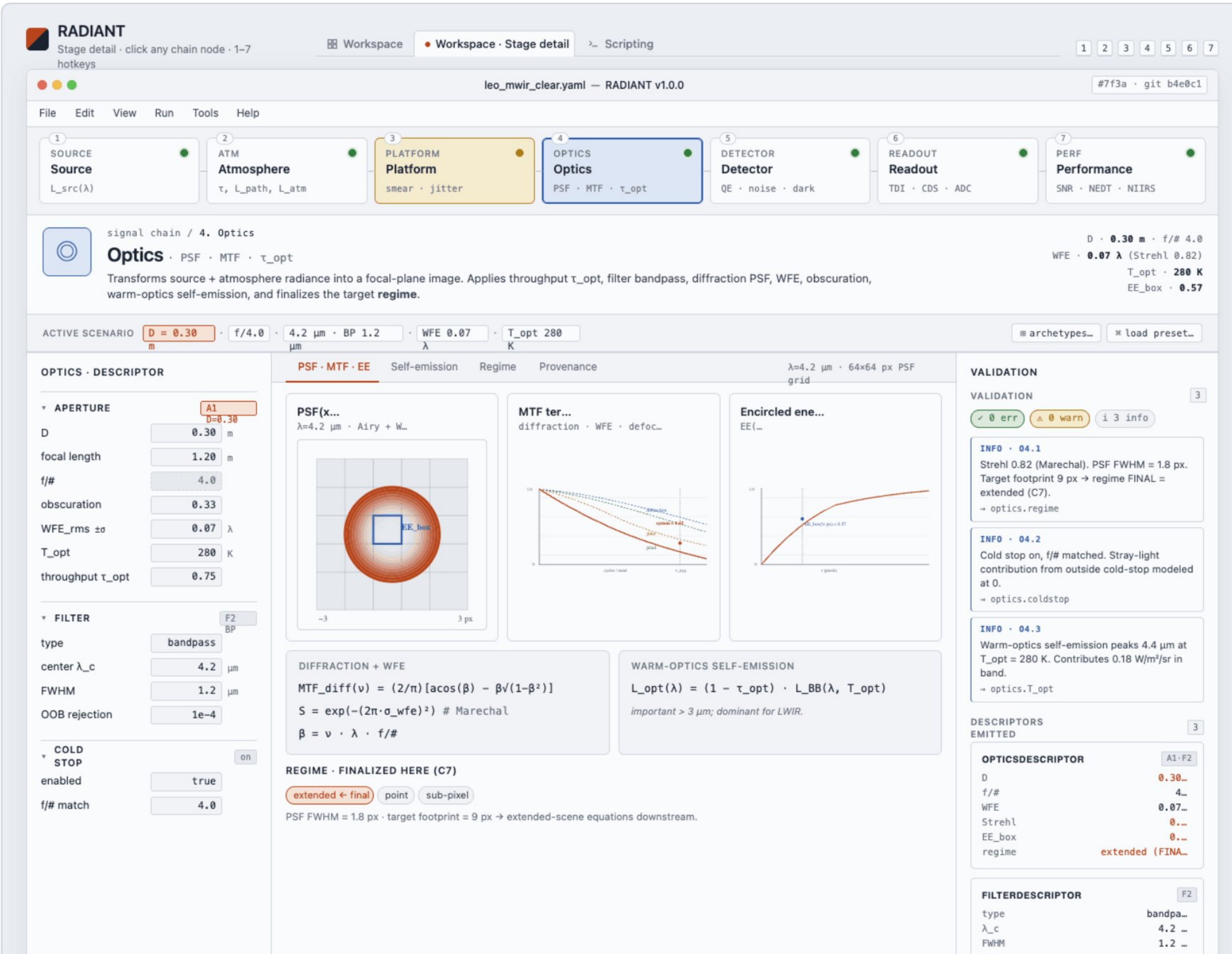
Platform

Line-of-sight motion — along-track smear, Gaussian jitter, and TDI mis-registration as MTF multipliers.



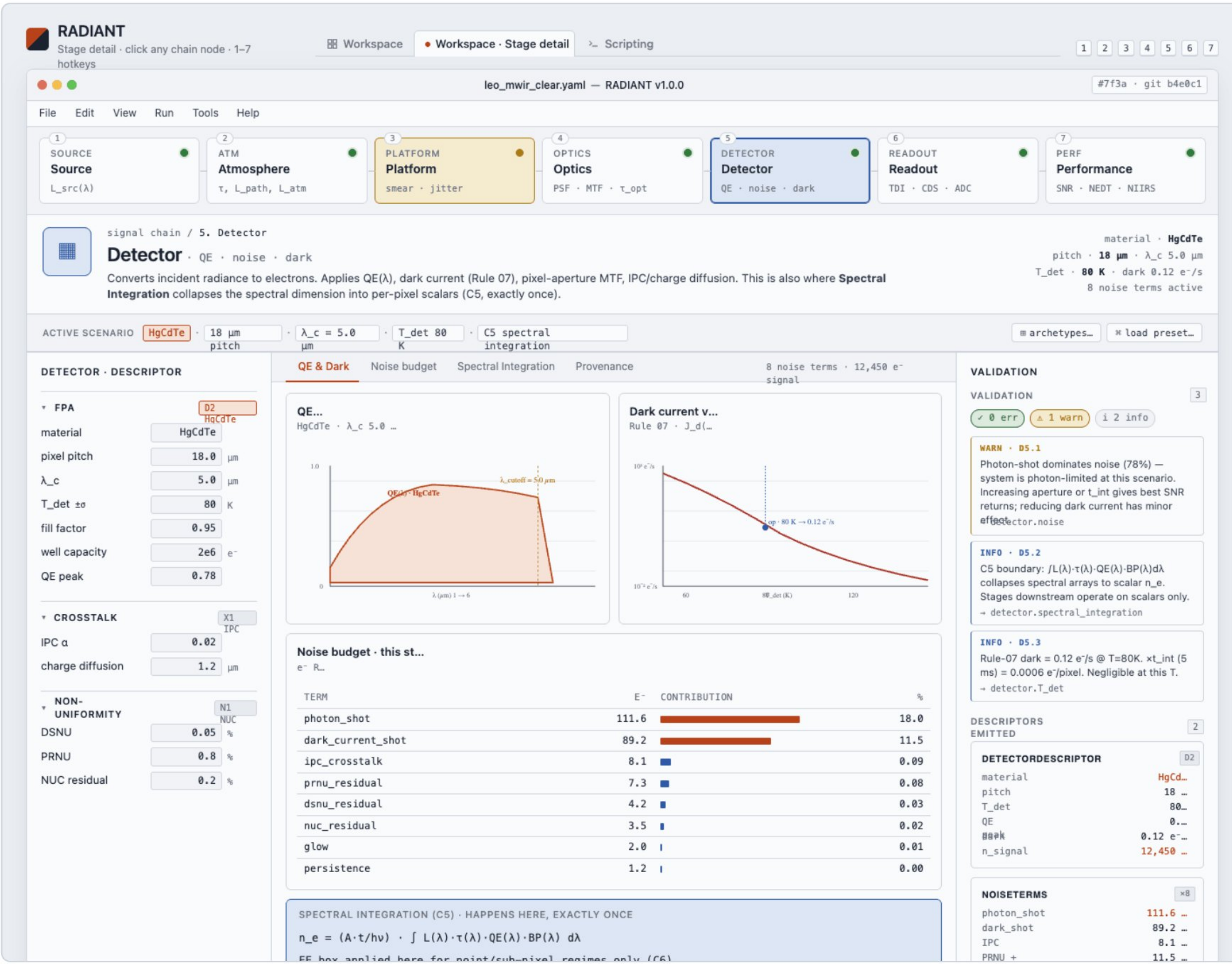
Optics

Diffraction PSF, MTF terms and encircled energy; warm-optics self-emission; filter bandpass; regime finalized (C7).



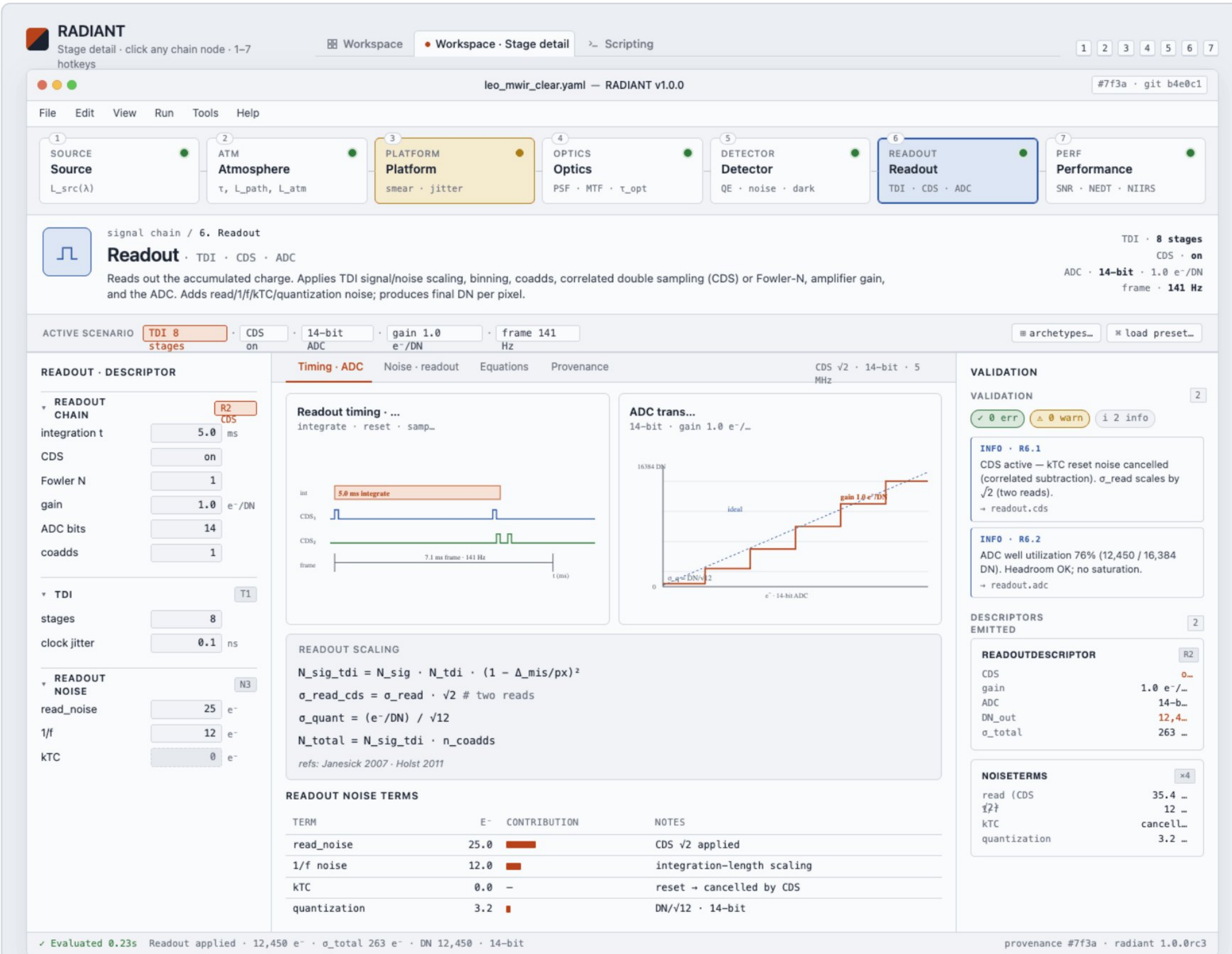
Detector

Converts radiance to electrons — QE(λ), Rule-07 dark current, eight-term noise budget, and the spectral-integration boundary (C5).



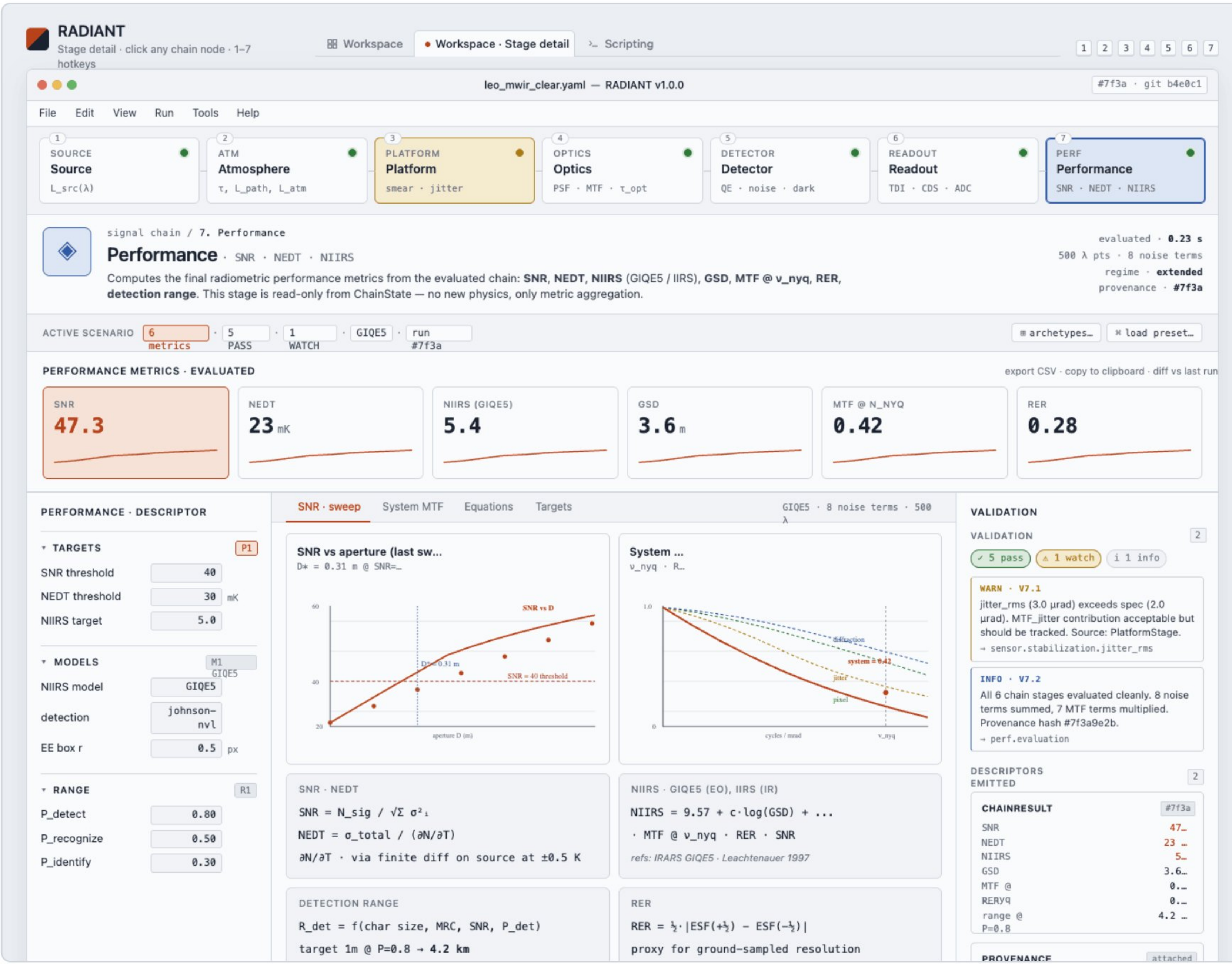
Readout

TDI / CDS timing and ADC transfer — read, 1/f, kTC and quantization noise, producing digitized counts per pixel.



Performance

SNR, NEDT, NIIRS (GIQE5/IIRS), GSD, MTF @ Nyquist, RER and detection range, with a pass/fail matrix and provenance manifest.



Scripting

The Python trade-study workspace — cell editor, live ChainState explorer, figures dock, and console, sharing the visual Workspace backend.